

Poornachandra G.

Lead Bioinformatics Scientist | Precision Oncology | NGS Architecture & Agentic AI Systems

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PROFESSIONAL SUMMARY

Bioinformatics Lead Scientist with 10+ years of experience in precision medicine in oncology. Expert in designing, building and productionizing end-to-end NGS and multi-omics pipelines for clinical workflows. Conceived and built the company's cross-functional Bioinformatics stack from scratch, delivering reproducible, scalable, and regulatory-ready solutions that integrate NGS analysis, clinical variant interpretation, and data-driven insights. Active builder of AI powered tools spanning multi-agent systems, full-stack applications, and developer tooling.

SELECTED HIGHLIGHTS & IMPACT

- Built and productionized end-to-end NGS pipelines (WGS/WES/TES, RNA-Seq, CNV, SV) used in clinical and research workflows; reduced sample turnaround time through automated QC and reporting.
- Architected variant annotation and prioritization workflows (ACMG/AMP/ASCO/CAP) integrating multiple public and proprietary annotations to enable clinical decision support.
- Led ML initiatives for biomarker discovery and patient stratification: feature engineering from multi-omics inputs, model validation, and deployment in production environments.
- Mentored and grew a bioinformatics team; led technical hiring, performance reviews and cross-functional training.
- Independently built 12+ independent AI projects spanning LLM tooling, multi-agent frameworks, clinical AI apps, and developer infrastructure, demonstrating end-to-end product ownership beyond core role.
- Delivered internal web applications and dashboards (Streamlit, Gradio, FastAPI) to deploy AI-based applications.

SKILLS & TECHNOLOGIES

NGS & Clinical-omics	WGS • WES • TES • RNA-Seq • scRNA-Seq • Liquid Biopsy • Pharmacogenomics • Companion Diagnostics • Variant Interpretation (ACMG/AMP/ASCO/CAP) • MSI • HRD • TMB • CNV • SV
Bioinformatics & Pipeline Engineering	GATK • Sentieon • DRAGEN • NVIDIA Parabricks • FreeBayes • DeepVariant • DeepSomatic • PureCN • CNVKit • VEP • ANNOVAR • Nextflow • QC Automation • Benchmarking • Reproducibility • Kallisto • Salmon • DESeq2 • edgeR • Seurat • Scanpy • GSEA
Machine Learning & AI	Supervised/Unsupervised Learning • Feature Engineering • LightGBM • XGBoost • PyTorch • TensorFlow • scikit-learn • PyCaret • AutoML • PySpark • SHAP • scipy • pandas • NumPy • R
LLMs & Agentic System	LangChain • LangGraph • Ollama • OpenAI/Anthropic/Gemini APIs • RAG Pipelines • Multi-Agent Systems • Prompt Engineering • Fine-Tuning • Quantization • LoRA/QLoRA • LiteLLM • MCP • W&B
Programming & Engineering	Python • Bash • R • FastAPI • SQLite • REST APIs • Uvicorn
Cloud, Platforms & Infrastructure	AWS (EC2, S3, Batch, HealthOmics) • DNAnexus • BaseSpace • Galaxy • GenePattern • Terra/FireCloud • Illumina Connected Analytics • Docker • Git • CI/CD

	• Conda/Mamba • Jupyter • HPC • MLOps
NLP, OCR & Document AI	Hugging Face Transformers • LayoutLMv3 • BioBERT • scispaCy • NLTK • Tesseract • EasyOCR • PyMuPDF • Docling • unstructured.io • OpenCV
Visualization & UI	Streamlit • Gradio • Plotly • Matplotlib • Seaborn
GenAI Developer Tool	Claude Code • Antigravity • Codex • Cursor • GitHub Copilot • Windsurf • Continue • Kiro • Cline • OpenCode • OpenClaw • Replit • Aider • Goose
Leadership & Delivery	Team Building • Hiring • Mentoring • Technical Roadmapping • Stakeholder Communication • HIPAA Compliance • Agile • Scrum • Risk Management • JIRA • Trello

PROFESSIONAL EXPERIENCE

Cellworks - Lead Bioinformatician

2024 - Present

- Led the Bioinformatics R&D team - defined the technical roadmap, recruited and mentored scientists, and drove project prioritization and delivery.
- Designed structured upskilling sessions leveraging supplemental projects beyond core deliverables, improving engagement, critical thinking, and problem-solving.
- Designed and implemented input ingestion, monitoring, and reporting frameworks that reduced manual review time by 20%.
- Built multi-stage annotation and prioritization pipelines applying AMP/ASCO/CAP guidelines across public and proprietary annotation sources.
- Partnered with executive leadership to align strategy and communicate progress, while providing team members with the broader context needed to execute effectively.
- Established risk management practices for bioinformatics pipelines - identified potential failure points, data-quality risks, and downstream repercussions, and implemented mitigation strategies.

Cellworks - Senior Scientist, Reporting Manager

2021 - 2024

- Developed workflows for MSI, HRD, mutation signatures, TMB, and indel burden.
- Built high-throughput variant calling stacks on AWS following GATK best practices, including mutation and CNV calling, and biomarker extraction.
- Built and led the bioinformatics team: recruited, trained, mentored, and reviewed the work of 4 members; created custom training protocols that accelerated onboarding and competency development.
- Designed input-agnostic workflows supporting multiple formats (FASTQ, BAM, VCF, structured reports) across diverse vendors (4BC, Tempus, Caris, Foundation Medicine, Alacris, etc.).
- Built cancer driver and CNV vs. mRNA databases for 30+ cancer types using public datasets/tools.
- Developed automated multi-omics interpretation pipelines for high-impact clinical trial datasets from Takeda and Genentech.
- Created an in-house VUS reclassification pipeline using an Ensembl-based approach, improving classification accuracy by >40%.

Cellworks - Senior Scientist, Bioinformatics

2018 - 2021

- Developed and benchmarked secondary analysis pipelines (Illumina DRAGEN, NVIDIA Parabricks, Sentieon), optimized for accuracy and turnaround time.

- Led custom panel design for liquid-biopsy samples; collaborated with Cleveland Clinic Florida to build an ML pipeline using longitudinal genomic data to predict early relapse in Melanoma.
- Built a framework to process pediatric cancer cases; served as primary owner for pediatric hematological cancers.
- Designed and implemented a Panel of Normals (PoN) workflow for CNV and mutation calling using ~100 cases, improving support for tumor-only analyses.
- Created multiple tertiary analysis workflows (GSEA, pathway enrichment, PCA, statistical models, ML pipelines) for biomarker discovery and treatment-response characterization.
- Developed multi-omics interpretation pipelines for the Foundation Medicine dataset (18,000+ cases).
- Compiled a comprehensive assay/test inventory across 88 cancers based on NCCN, FDA, and other clinical guidelines.

Cellworks - Scientist, Clinical Translation

2017 - 2018

- Curated rationale libraries to support therapy predictions, reducing turnaround time for live case analyses by >40%.
- Analyzed 200+ cancer cases, gaining deep expertise in target selection using cancer and drug networks in precision oncology.
- Conducted benchmarking of sequencing pipelines (DNAnexus, BeOne, MedGenome) to establish confidence and reproducibility.
- Developed an in-house variant annotation and filtering pipeline, lowering processing costs and enabling greater customization.
- Implemented a custom variant-tier classification system to improve prioritization and enhance interpretability of genomic reports.

Cellworks - Associate Scientist

2015 - 2017

- Processed, interpreted, and harmonized multi-omics outputs (karyotyping, aCGH, IHC, mutations) for >500 cases, translating insights into automated data-harmonization and ETL pipelines.
- Curated and automated retrospective in-house dataset creation by integrating patient and cell-line data from >10,000 GDSC, TCGA, and published studies.
- Standardized genomic data interpretation from external partners (Columbia University, University of Florida, Cedars-Sinai, Mayo Clinic, Washington University, MMRF-Michigan).
- Automated report-generation workflows using reproducible pipelines and templates, reducing preparation time from ~15 hours to just minutes per week.

AI & INDEPENDENT PROJECTS

Clinical & Bioinformatics AI

- **Evaline:** Interactive chatbot combining a fine-tuned open-source LLM with a RAG pipeline to deliver guideline-based treatment recommendations and patient-friendly explanations.
- **PathoParse:** LLM-assisted and rule-based pipelines to extract structured clinical metadata from unstructured pathology reports for downstream analysis.
- **SQL Chat:** Multi-agent Streamlit demo that translates natural language clinical queries to SQL, runs statistical tests, and returns interpretable Markdown summaries for non-technical users.
- **VarEffex:** Multi-agent framework that mines literature to extract variant-function associations.

Agentic & LLM Tooling

- **AnalomX**: Local-first agentic exploratory data analysis framework for tabular datasets (CSV/TSV). Multi-agent workflow with automatic profiling, code-driven analysis, critique loop, and report generation (Markdown/HTML/PDF/Notebook). FastAPI backend with browser UI.
- **DigiTeam**: Framework for building a digital version of bioinformatics team. Captures team context, skills, SOPs, intake workflows, MCP resource access and guardrails - designed for future CrewAI-style agent orchestration.
- **MemorAI**: Personal AI context database stored as plain Markdown for Obsidian. Captures professional and personal context as linked atomic notes so AI systems can understand user context without repeated re-explanation.

Full-Stack Applications

- **JobAlder**: Production-ready AI job application assistant. Builds master professional profiles, analyzes job descriptions, performs gap analysis, generates ATS-optimized resumes, and runs mock interviews. Stack: FastAPI + Next.js+ LangChain + SQLite.
- **HealthForge**: AI-assisted health planning platform. Manages health profiles, generates AI-based health plans, supports AI chat in profile context, and schedules WhatsApp reminders. Stack: Next.js, SQLite, Baileys, node-cron.
- **LinkedDjin**: Comprehensive LinkedIn content generation suite. Features interactive CLI, non-interactive CLI for automation, FastAPI web GUI, cron-based scheduling, draft library, LinkedIn-ready Unicode formatting, and auto-posting integration.

Developer Tooling & Infrastructure

- **Inferverse**: LiteLLM configuration generator and local AI proxy. Automates discovery and categorization of models across NVIDIA NIM, Mistral, Groq, OpenRouter, Ollama, and PublicAI. Includes a Claude Code bridge for Claude Code and VS Code's Continue integration.
- **Tragic (TRAdE loGIC)**: Portfolio stop-loss and re-entry dashboard for Indian investors trading US stocks. Live Yahoo Finance prices, 4-tier re-entry levels, STCG/LTCG tax calculator, price charts, and MCP server for Claude Desktop integration.

EDUCATION

B.Tech, Biotechnology - JNTUA CEP

2015

SELECTED PUBLICATIONS & ABSTRACTS

Peer-Reviewed Journals

- Castro M, ..., **Poornachandra G**, et al. Combination chemotherapy versus temozolomide for patients with methylated MGMT glioblastoma: computational biological modeling predictions. J Neurooncol. 2021;153(3):393-402.
- Castro M, ..., **Poornachandra G**, et al. Cellworks Omics Biology Model (CBM) to predict therapy response and identify novel biomarkers for 5FU-based combination therapy in gastric cancer patients. J Clin Oncol. 2021;39(15_suppl):e16024.

Conference Abstracts (Blood, ASCO, JCO)

- Marcucci G, ..., **Poornachandra G**, et al. Biosimulation using CBM identifies novel biomarkers to inform MEC-based combination therapy in refractory & relapsed AML patients. Blood. 2021;138(Suppl_1).
- Castro M, ..., **Poornachandra G**, et al. Biosimulation using CBM-based assessment of cytarabine and anthracycline response in AML. Blood. 2021;138(Suppl_1).
- Howard SC, ..., **Poornachandra G**, et al. Therapy biosimulation using CBM is predictive of individual AML patient probability of CR and OS: MyCare-023. Blood. 2021;138(Suppl_1).
- Castro M, ..., **Poornachandra G**, et al. Monosomy 7 and co-occurrent genomic aberrations determine chemotherapy response in AML: MyCare-021-04. Blood. 2020;136(Suppl_1).
- Howard SC, ..., **Poornachandra G**, et al. Predicting resistance to ATO and ATRA in APL patients with PML-RARA fusions: MyCare-021-01. Blood. 2020;136(Suppl_1).
- Ahluwalia MS, ..., **Poornachandra G**, et al. Superior therapy response predictions for LGG using Cellworks Singula: MyCare-009-04. J Clin Oncol. 2020;38(15_suppl):e13541.
- Vali S, ..., **Poornachandra G**, et al. Analysis of evolving MDS/AML clones to identify resistance mechanisms: iCare1. Blood. 2018;132(Suppl_1).
- Vij R, ..., **Poornachandra G**, et al. Predicting response to azacitidine or lenalidomide/azacitidine combination in MDS and AML using CBM. Blood. 2018;132(Suppl_1).